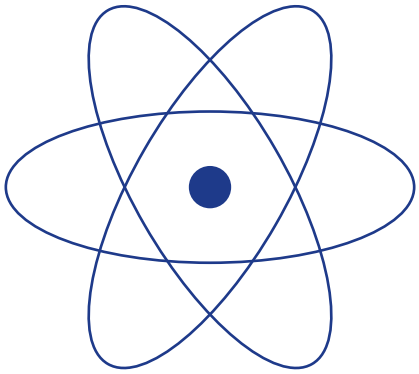
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



# Fluids

## Olympiad Solutions Manual

Olympiad Training • Mechanics

SPhOH Team

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CHAPTER 1

# Fluids Solutions

## 1.1 Units and Density Conversion

### SOLUTION 1

Density Unit Conversion I

**1. Conversion Factors**
Density is mass per unit volume ($\rho = m/V$). The relationship between SI units ($\mathrm{kg/m^3}$) and CGS units ($\mathrm{g/cm^3}$) is:

$$1\ \mathrm{g/cm^3} = \frac{10^{-3}\ \mathrm{kg}}{(10^{-2}\ \mathrm{m})^3} = \frac{10^{-3}\ \mathrm{kg}}{10^{-6}\ \mathrm{m^3}} = 1000\ \mathrm{kg/m^3}$$

**2. Calculation**
To convert from $\mathrm{kg/m^3}$ to $\mathrm{g/cm^3}$, we divide by 1000:

$$\rho = \frac{200\ \mathrm{kg/m^3}}{1000\ \mathrm{kg/m^3/(g/c^3m)}} = 0.2\ \mathrm{g/cm^3}$$

### SOLUTION 2

Density Unit Conversion II

**1. Given Parameters**
Let 1 R = $\rho_{\mathrm{oil}} = 600\ \mathrm{kg/m^3}$. Standard water density is $\rho_{\mathrm{water}} = 1000\ \mathrm{kg/m^3}$.
**2. Ratio Calculation**
To express water's density in terms of the new unit R, we compute the ratio:

$$\text{Density of water in R} = \frac{\rho_{\mathrm{water}}}{\rho_{\mathrm{oil}}} = \frac{1000\ \mathrm{kg/m^3}}{600\ \mathrm{kg/m^3}} = \frac{5}{3}\ \mathrm{R} \approx 1.67\ \mathrm{R}$$

## 1.2 Hydrostatic Pressure and Pascal's Principle

### SOLUTION 3

Pressure Under Heavy Metallic Cylinder

**1. Physical Parameters and Unit Conversions**

- Cylinder mass: $m = 80$ kg
- Base area: $A = 25\ \text{cm}^2 = 25 \times 10^{-4}\ \text{m}^2 = 2.5 \times 10^{-3}\ \text{m}^2$
- Gravitational acceleration: $g = 9.8\ \text{m/s}^2$

**2. Force and Pressure Calculation**
The normal force exerted on the ground equals the gravitational force on the cylinder:

$$F = mg = (80\ \text{kg})(9.8\ \text{m/s}^2) = 784\ \text{N}$$

The pressure $P$ exerted on the flat planar base is:

$$P = \frac{F}{A} = \frac{784\ \text{N}}{2.5 \times 10^{-3}\ \text{m}^2} = 313600\ \text{Pa} = 313.6\ \text{kPa}$$

*(Note: If taking $g = 10\ \text{m/s}^2$, $P = 320$ kPa.)*

## SOLUTION 4

Force Scaling in Hydraulic Press

**1. Pascal's Principle Formulation**
Pascal's Principle states that an external pressure applied to a confined fluid is transmitted equally to every point of the fluid. The pressure $P$ exerted by the small piston is:

$$P = \frac{f}{a}$$

**2. Sustained Force**
To maintain static equilibrium, the pressure at the larger piston must be equal:

$$P = \frac{F}{A} \implies \frac{f}{a} = \frac{F}{A}$$

Solving explicitly for $F$:

$$F = f\left(\frac{A}{a}\right)$$

## SOLUTION 5

Circular Hydraulic Lift

**1. Geometry and Area Ratios**
For circular pistons with diameters $d_1 = 5.0$ cm and $d_2 = 25.0$ cm, their cross-sectional areas are:

$$A_1 = \frac{\pi d_1^2}{4}, \quad A_2 = \frac{\pi d_2^2}{4} \implies \frac{A_2}{A_1} = \left(\frac{d_2}{d_1}\right)^2 = \left(\frac{25.0}{5.0}\right)^2 = 25$$

**2. Force Required for Static Equilibrium**
The required force $F_2$ to lift the vehicle of mass $m = 1500$ kg under $g = 10$ m/s$^2$ is:

$$F_2 = mg = (1500 \text{ kg})(10 \text{ m/s}^2) = 15000 \text{ N}$$

Applying Pascal's Principle:

$$F_1 = F_2 \left(\frac{A_1}{A_2}\right) = 15000 \text{ N} \times \frac{1}{25} = 600 \text{ N}$$

## SOLUTION 6

Hydrostatic Pressure Difference and Depth

**1. Part (a): Pressure Comparison**
The fundamental hydrostatic equation is $\Delta P = \rho g \Delta h$. Because hydrostatic pressure increases monotonically with vertical depth, Point B (being lower in the water column) is at a higher pressure than Point A.
**2. Part (b): Vertical Distance Separation**
Given $\Delta P = 500$ Pa, $\rho = 1000$ kg/m$^3$, and $g = 10$ m/s$^2$:

$$\Delta h = \frac{\Delta P}{\rho g} = \frac{500 \text{ Pa}}{(1000 \text{ kg/m}^3)(10 \text{ m/s}^2)} = 0.05 \text{ m} = 5.0 \text{ cm}$$

**3. Part (c): Absolute Pressure at Point B**
The gauge pressure at Point A (at depth $h_A = 0.10$ m) is:

$$P_A = \rho g h_A = (1000)(10)(0.10) = 1000 \text{ Pa}$$

Since Point B is lower by $\Delta h = 0.05$ m (depth $h_B = 0.15$ m):

$$P_B = P_A + \Delta P = 1000 \text{ Pa} + 500 \text{ Pa} = 1500 \text{ Pa}$$

## SOLUTION 7

Pressure Exerted by Chair Legs

**1. Force Distribution**
The total weight of the man and chair is $W = mg$. Assuming symmetric mass distribution,

the load is divided equally among the 4 legs:

$$F_{\text{leg}} = \frac{mg}{4}$$

**2. Contact Area and Pressure**
Each circular leg has radius $r$, giving a bottom contact area $A_{\text{leg}} = \pi r^2$. The pressure exerted by each individual leg on the floor is:

$$P = \frac{F_{\text{leg}}}{A_{\text{leg}}} = \frac{mg/4}{\pi r^2} = \frac{mg}{4\pi r^2}$$

## SOLUTION 8

**Adding Water over Mercury to Double Base Pressure**

**1. Initial Base Pressure**
Let $h_m = 8.00\text{ cm} = 0.08\text{ m}$ be the initial height of mercury ($\rho_m = 13.6\text{ g/cm}^3$). The initial hydrostatic pressure at the bottom is:

$$P_{\text{initial}} = \rho_m g h_m$$

**2. Hydrostatic Requirement for Doubling Pressure**
When water ($\rho_w = 1.00\text{ g/cm}^3$) is poured on top to height $h_w$, the new base pressure is:

$$P_{\text{final}} = P_{\text{initial}} + \rho_w g h_w = 2P_{\text{initial}} \implies \rho_w g h_w = \rho_m g h_m$$

Canceling $g$:

$$h_w = \left(\frac{\rho_m}{\rho_w}\right) h_m = \left(\frac{13.6\text{ g/cm}^3}{1.00\text{ g/cm}^3}\right)(8.00\text{ cm}) = 108.8\text{ cm}$$

**3. Required Water Volume**
Given the cylinder cross-sectional area $A = 12.0\text{ cm}^2$:

$$V_w = Ah_w = (12.0\text{ cm}^2)(108.8\text{ cm}) = 1305.6\text{ cm}^3 = 1.3056\text{ L}$$

## SOLUTION 9

**Critical Water Height in Sealed U-Tube**

**1. Maximum Gauge Pressure at the Seal**
The seal pops off when the force exceeds $F_{\text{max}} = 9.80\text{ N}$. Given cross-sectional area $A = 5.00\text{ cm}^2 = 5.00 \times 10^{-4}\text{ m}^2$, the maximum tolerable gauge pressure is:

$$P_{\text{gauge, max}} = \frac{F_{\text{max}}}{A} = \frac{9.80\text{ N}}{5.00 \times 10^{-4}\text{ m}^2} = 19600\text{ Pa}$$

**2. Water Level Difference $\Delta h$**

The pressure difference between the water level in the open arm and the sealed short arm is generated by the height difference $\Delta h$:

$$P_{\text{gauge, max}} = \rho_w g \Delta h \implies \Delta h = \frac{19600 \text{ Pa}}{(1000 \text{ kg/m}^3)(9.80 \text{ m/s}^2)} = 2.00 \text{ m}$$

**3. Total Water Height in Long Arm**
Since the short arm is completely full to height $d = 0.800$ m, the total height of water column in the long arm is:

$$H_{\text{total}} = d + \Delta h = 0.800 \text{ m} + 2.00 \text{ m} = 2.80 \text{ m}$$

## 1.3 Buoyancy and Floating Bodies

### SOLUTION 10

Floating Object under Varied Gravity

**1. Equilibrium Condition**
For a floating object of mass $m$ and total volume $V$ in a liquid of density $\rho_{\text{fluid}}$:

$$F_B = F_g \implies \rho_{\text{fluid}} V_{\text{sub}} g = \rho_{\text{object}} V g$$

**2. Independence from $g$**
Dividing both sides by $g$ (assuming $g > 0$):

$$\frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$$

The submerged volume fraction depends strictly on the ratio of densities. Doubling the planetary gravitational acceleration $2g$ scales both the downward weight and the upward buoyant force by a factor of 2 equally. Thus, the submerged volume fraction remains exactly $\frac{1}{4}$ (or 25%).

### SOLUTION 11

Equilibrium Submerged Fraction

**1. Static Balance**
When an object floats with $V_{\text{sub}} = 0.5 V_{\text{tot}}$ in water ($\rho_w$):

$$F_B = W \implies \rho_w V_{\text{sub}} g = \rho_{\text{obj}} V_{\text{tot}} g$$

**2. Density Ratio**

Canceling $g$ and substituting $V_{\text{sub}}/V_{\text{tot}} = 0.5$:

$$\frac{\rho_{\text{obj}}}{\rho_w} = \frac{V_{\text{sub}}}{V_{\text{tot}}} = 0.5$$

The object's density is exactly half the density of water (500 kg/m$^3$).

## SOLUTION 12

Emergent Height of Floating Cylinder

**1. Submerged Height Calculation**
For a uniform vertical cylinder of height $h_{\text{tot}} = 10$ cm and density $\rho_{\text{cyl}} = 0.3$ g/cm$^3$ floating in water ($\rho_w = 1.0$ g/cm$^3$):

$$\rho_w A h_{\text{sub}} g = \rho_{\text{cyl}} A h_{\text{tot}} g \implies h_{\text{sub}} = h_{\text{tot}} \left(\frac{\rho_{\text{cyl}}}{\rho_w}\right) = (10\text{ cm}) \left(\frac{0.3}{1.0}\right) = 3.0\text{ cm}$$

**2. Emergent Height**
The height extending above the water line is:

$$h_{\text{above}} = h_{\text{tot}} - h_{\text{sub}} = 10\text{ cm} - 3.0\text{ cm} = 7.0\text{ cm}$$

## SOLUTION 13

Submerged Ratio in Different Liquids

**1. Force Balance in Both Liquids**
The total mass $m$ and cross-sectional area $A$ of the upright box are constant. In water:

$$W = \rho_{\text{water}} A h_w g$$

In oil:

$$W = \rho_{\text{oil}} A h_o g$$

**2. Ratio Derivation**
Equating the two expressions for weight $W$:

$$\rho_{\text{water}} A h_w g = \rho_{\text{oil}} A h_o g \implies \frac{h_w}{h_o} = \frac{\rho_{\text{oil}}}{\rho_{\text{water}}}$$

## SOLUTION 14

Cavity Volume in Cast Metal

**1. Apparent Weight and Buoyant Force**
Given $W_{\text{air}} = 6000$ N and $W_{\text{water}} = 4000$ N under $g = 9.8$ m/s$^2$:

$$F_B = W_{\text{air}} - W_{\text{water}} = 6000 \text{ N} - 4000 \text{ N} = 2000 \text{ N}$$

**2. External Outer Volume**
By Archimedes' Principle, $F_B = \rho_w g V_{\text{ext}}$:

$$V_{\text{ext}} = \frac{F_B}{\rho_w g} = \frac{2000 \text{ N}}{(1000 \text{ kg/m}^3)(9.8 \text{ m/s}^2)} \approx 0.20408 \text{ m}^3$$

**3. Solid Iron Volume and Cavity Calculation**
The volume occupied purely by solid iron ($\rho_{\text{iron}} = 7870$ kg/m$^3$) is:

$$V_{\text{iron}} = \frac{W_{\text{air}}}{\rho_{\text{iron}} g} = \frac{6000 \text{ N}}{(7870 \text{ kg/m}^3)(9.8 \text{ m/s}^2)} \approx 0.07779 \text{ m}^3$$

The total cavity volume is:

$$V_{\text{cavity}} = V_{\text{ext}} - V_{\text{iron}} = 0.20408 \text{ m}^3 - 0.07779 \text{ m}^3 = 0.12629 \text{ m}^3 \approx 0.126 \text{ m}^3 \text{ (126.3 L)}$$

## SOLUTION 15 — Tension in Submerged Metal Ball

**1. Force Analysis**
The ball is suspended by a string, submerged in water. Downward forces are Weight $W$, upward forces are Tension $T$ and Buoyant Force $F_B$:

$$T + F_B = W \implies T = W - F_B$$

**2. Numerical Evaluation**
Volume $V = 1000$ cm$^3 = 10^{-3}$ m$^3$, $\rho_m = 2500$ kg/m$^3$, $\rho_w = 1000$ kg/m$^3$, $g = 10$ m/s$^2$.

$$W = \rho_m V g = (2500)(10^{-3})(10) = 25 \text{ N}$$

$$F_B = \rho_w V g = (1000)(10^{-3})(10) = 10 \text{ N}$$

$$T = 25 \text{ N} - 10 \text{ N} = 15 \text{ N}$$

## SOLUTION 16 — Density of Floating Hollow Cylinder

**1. Volume Definitions**
Let $L$ be the length of the cylinder.

- Total outer cross-sectional volume: $V_{\text{outer}} = \pi R_{\text{outer}}^2 L$

- Material cross-sectional volume: $V_{\text{mat}} = \pi(R_{\text{outer}}^2 - R_{\text{inner}}^2)L$
- Submerged volume (half of total volume): $V_{\text{sub}} = \frac{1}{2}\pi R_{\text{outer}}^2 L$

**2. Equilibrium Balance**
Equating buoyant force to total weight:

$$\rho_{\text{liquid}} V_{\text{sub}} g = \rho_{\text{mat}} V_{\text{mat}} g \implies \rho_{\text{liquid}} \left(\frac{1}{2}\pi R_{\text{outer}}^2 L\right) = \rho_{\text{mat}} \left[\pi(R_{\text{outer}}^2 - R_{\text{inner}}^2)L\right]$$

**3. Material Density Calculation**
Given $R_{\text{outer}} = 8.0$ cm, $R_{\text{inner}} = 5.0$ cm, $\rho_{\text{liquid}} = 0.65$ g/cm$^3$:

$$\rho_{\text{mat}} = \rho_{\text{liquid}} \frac{0.5R_{\text{outer}}^2}{R_{\text{outer}}^2 - R_{\text{inner}}^2} = 0.65 \times \frac{0.5(64)}{64 - 25} = 0.65 \times \frac{32}{39} = \frac{20.8}{39} \approx 0.533 \text{ g/cm}^3$$

## SOLUTION 17

Log Raft Capacity

**1. Single Log Parameters**
Radius $r = 0.15$ m, length $L = 1.80$ m, density $\rho_{\log} = 800$ kg/m$^3$. Volume of one log:

$$V = \pi r^2 L = \pi(0.15 \text{ m})^2(1.80 \text{ m}) \approx 0.12723 \text{ m}^3$$

**2. Payload Capacity per Log**
When fully submerged in fresh water ($\rho_w = 1000$ kg/m$^3$ under $g = 9.80$ m/s$^2$), the net upward force per log available to support children is:

$$F_{\text{net, log}} = (\rho_w - \rho_{\log})V g = (1000 - 800)(0.12723)(9.80) \approx 249.38 \text{ N}$$

**3. Minimum Log Count $N$**
Total weight of 3 children: $W_{\text{total}} = 3 \times 356$ N $= 1068$ N.

$$N \geq \frac{1068 \text{ N}}{249.38 \text{ N/log}} \approx 4.28$$

Rounding up to the nearest integer, a minimum of $N = 5$ logs are required.

## SOLUTION 18

Floating Body at Two-Fluid Interface

**1. Variable Definition**
Let $f$ be the fraction of the box's volume submerged in fluid A ($\rho_A = 1200$ kg/m$^3$). The remaining fraction $(1 - f)$ is submerged in fluid B ($\rho_B = 800$ kg/m$^3$).

**2. Buoyancy Equilibrium**
Total buoyant force matches the weight of the box ($\rho_{\text{box}} = 900\ \text{kg/m}^3$):

$$\rho_A(fV)g + \rho_B(1-f)Vg = \rho_{\text{box}}Vg$$

Canceling $Vg$:

$$f\rho_A + (1-f)\rho_B = \rho_{\text{box}}$$

**3. Solving for $f$**
Substitute density values:

$$1200f + 800(1-f) = 900 \implies 400f = 100 \implies f = 0.25 = 25\%$$

## SOLUTION 19 — Dual Submersion Apparent Weight

**1. Governing Equations**
Let $W$ be the true weight of the rock in air and $V$ its volume. In water ($\rho_w = 1.00\ \text{g/cm}^3 = 1000\ \text{kg/m}^3$):

$$W - \rho_w Vg = 1.20\ \text{N} \quad \text{— (1)}$$

In ethanol ($\rho_e = 0.79\ \text{g/cm}^3 = 790\ \text{kg/m}^3$):

$$W - \rho_e Vg = 1.60\ \text{N} \quad \text{— (2)}$$

**2. Eliminating Volume $V$**
Subtracting Equation (1) from Equation (2):

$$(\rho_w - \rho_e)Vg = 1.60 - 1.20 = 0.40\ \text{N}$$

Since $\rho_w - \rho_e = (1.00 - 0.79)\text{g/cm}^3 = 0.21\text{g/cm}^3$:

$$\rho_w Vg = \frac{\rho_w}{\rho_w - \rho_e}(0.40\ \text{N}) = \frac{1.00}{0.21}(0.40\ \text{N}) \approx 1.9048\ \text{N}$$

**3. Calculating Weight $W$**
Substituting back into Equation (1):

$$W = 1.20\ \text{N} + 1.9048\ \text{N} = 3.1048\ \text{N} \approx 3.10\ \text{N}$$

## SOLUTION 20 — Wood Block Weighted with Lead

**1. Wood Block Volume**

Given mass $m_w = 3.67$ kg and density $\rho_w = 600$ kg/m$^3$:

$$V_w = \frac{m_w}{\rho_w} = \frac{3.67 \text{ kg}}{600 \text{ kg/m}^3} = 6.1167 \times 10^{-3} \text{ m}^3$$

**2. Part (a): Lead on Top (in Air)**
Only $0.900V_w$ of the wood provides buoyant force in water ($\rho_{\text{water}} = 1000$ kg/m$^3$):

$$\rho_{\text{water}}(0.900V_w)g = (m_w + m_L)g$$

Substituting values:

$$m_w + m_L = (1000 \text{ kg/m}^3)(0.900 \times 6.1167 \times 10^{-3} \text{ m}^3) = 5.505 \text{ kg}$$

$$m_L = 5.505 \text{ kg} - 3.67 \text{ kg} = 1.835 \text{ kg} \approx 1.84 \text{ kg}$$

**3. Part (b): Lead on Bottom (Submerged)**
Both the wood ($0.900V_w$) and the submerged lead ($V_L = m_L/\rho_L$, where $\rho_L = 1.14 \times 10^4$ kg/m$^3$) displace water:

$$\rho_{\text{water}} \left(0.900V_w + \frac{m_L}{\rho_L}\right) g = (m_w + m_L)g$$

Substitute $\rho_{\text{water}}(0.900V_w) = 5.505$ kg:

$$5.505 + m_L \left(\frac{\rho_{\text{water}}}{\rho_L}\right) = 3.67 + m_L$$

$$5.505 - 3.67 = m_L \left(1 - \frac{1000}{11400}\right) \implies 1.835 = m_L \left(1 - \frac{1}{11.4}\right) = m_L \left(\frac{10.4}{11.4}\right)$$

Solving for $m_L$:

$$m_L = 1.835 \times \frac{11.4}{10.4} \approx 2.011 \text{ kg} \approx 2.01 \text{ kg}$$